

Session 17

Dart Functions Assignment

Objective: Practice creating reusable functions, arrow functions, validation logic, and code refactoring in Dart.

Task 1: Validate Username

Description: Create a function **isValidUsername** that checks if a username is at least 6 characters long and contains no spaces.

```
bool isValidUsername(String username) {  
    return username.length >= 6 && !username.contains(' ');  
}  
  
void main() {  
    print(isValidUsername("jainish"));  
    print(isValidUsername("ab cd"));  
}
```

Expected Output:

true
false

Task 2: Calculate Discounted Price

Description: Create a reusable function that calculates the final price after applying a discount.

```
double calculateDiscountedPrice(  
    double originalPrice,  
    double discountPercentage) {  
    return originalPrice -  
        (originalPrice * discountPercentage / 100);  
}  
  
void main() {  
    print(calculateDiscountedPrice(1500, 20));  
}
```

Expected Output:

1200.0

Task 3: Arrow Function

Description: Use an arrow function to format follower counts like YouTube.

```
String formatFollowers(int n) =>
    n >= 1000000
    ? "${(n/1000000).toFixed(1)}M"
    : n >= 1000
    ? "${(n/1000).toFixed(1)}K"
    : "$n";

void main() {
    print(formatFollowers(1500));
    print(formatFollowers(1200000));
    print(formatFollowers(950));
}
```

Expected Output:

1.5K
1.2M
950

Task 4: Validate Indian Phone Number

Description: Create a function to validate Indian mobile numbers.

```
bool validatePhoneNumber(String phone) {
    RegExp pattern = RegExp(r'^[6-9]\d{9}\$');
    return pattern.hasMatch(phone);
}

void main() {
    print(validatePhoneNumber("9876543210"));
    print(validatePhoneNumber("1234567890"));
}
```

Expected Output:

true
false

Task 5: Refactor Using Function

Description: Extract the repeated service charge calculation into a reusable function.

```
double addServiceCharge(double price) {  
    return price + (price * 0.05);  
}  
  
void main() {  
    print(addServiceCharge(250));  
    print(addServiceCharge(450));  
    print(addServiceCharge(800));  
}
```

Expected Output:

262.5
472.5
840.0

Conclusion: This session demonstrated reusable Dart functions, arrow functions, validation using conditions and regular expressions, and refactoring repeated logic into clean, maintainable code.